

Build It in Detroit: A Parallelogram Design Project

Day 12 Enrichment · MathCraft · Grade 6 · Professor Xavier Honablue, M.Ed.

ENRICHMENT PROJECT

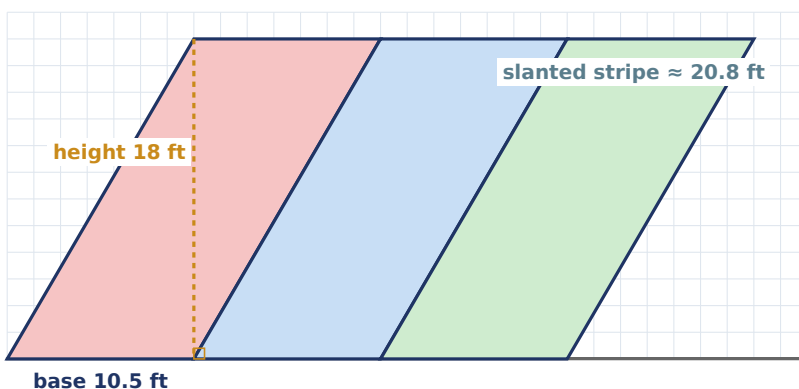
Materials: graph paper ($\frac{1}{4}$ -inch squares is best) · ruler · colored pencils · pencil & eraser. No calculators needed. **Standards:** 6.G.A.1 (area of parallelograms and composite shapes), 6.NS.B.3 (multiplying decimals), 6.G.A.3 (shapes on the coordinate plane). **Rule:** use your ruler for every straight line, and mark every height with a dotted line and a small right-angle box.

Why parallelograms matter in building design

Architects and engineers use parallelograms whenever they need shapes that **lean** but still **line up** in neat rows: angled parking stalls, diamond-shaped tile and brick patterns, roof panels, stair railings, and ramps. A **square** is a special parallelogram, too — which is why a baseball "diamond" counts. In this project you will be the designer for three Detroit-inspired jobs.

Job 1 · Design an Angled Parking Row (main task)

Many downtown lots use **angled parking** so drivers can pull in with one easy turn. Each angled stall is a parallelogram. The Indiana Department of Transportation design manual lists a 60° stall as **10.5 ft** wide along the aisle and **18.0 ft** deep (measured straight back from the aisle — that's the height). The painted stripe is slanted, so it is longer: about 20.8 ft.



Scale:

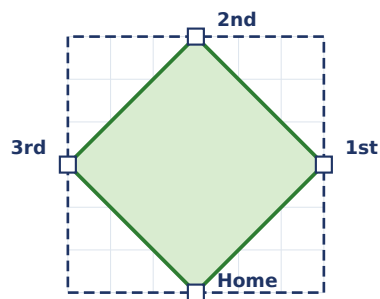
1 square = 1.5 ft

Stall: base 7 squares,
height 12 squares,
leans 7 squares.

1. On graph paper, copy the row of 3 stalls using the scale 1 square = 1.5 ft (base 7 squares, height 12 squares, each stall leans 7 squares). Color the stalls in 3 different colors.
2. Label one stall's base (10.5 ft), height (18 ft), and stripe (≈ 20.8 ft). Circle the two numbers that give the area.
3. Find the area of one stall: $10.5 \times 18 = \underline{\hspace{1cm}}$ sq ft. (Day 10 decimal skills!) Then find the area of all 3 stalls.
4. A straight-in (90°) stall is often 9 ft by 18 ft. Find its area. Which stall uses more pavement? How much more?
5. Distractor check: Jaylen says one stall is 10.5×20.8 sq ft. Explain his mistake in one sentence.

Job 2 • The Baseball Diamond (Comerica Park inspiration)

Comerica Park, home of the Detroit Tigers, has the same infield as every Major League field: the four bases form a **square with 90-ft sides**. Turned on its corner it looks like a "diamond," but it is still a square — and every square is a parallelogram (opposite sides are parallel).



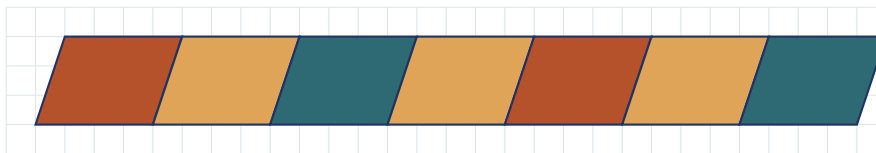
Corners: Home (3, 0), 1st (6, 3),
2nd (3, 6), 3rd (0, 3)

Dashed box = 6×6 grid squares
Cut off 4 corner triangles

6. Draw the diamond on graph paper using the corners shown. Color the infield green and the bases white.
7. Use the **box method**: 6×6 box minus the 4 corner triangles. What is the diamond's area in grid squares?
8. The real infield is 90 ft $\times$ 90 ft. Find its area in square feet.
9. Is a baseball diamond a parallelogram? Give two reasons.

Job 3 • An Art Deco Facade Band (Guardian Building inspiration)

Detroit's **Guardian Building** (1929, architect Wirt C. Rowland) is famous for colored brick, tile, and terra cotta set in bold **geometric patterns**. Design your own decorative band of leaning parallelogram tiles for a new Detroit building.



Each tile: base 4,
height 3, leans 1
(1 square = 1 ft)

10. Draw a band of 7 matching parallelogram tiles (base 4 squares, height 3 squares, each leaning 1 square). Use a repeating 4-color pattern.
11. Area of one tile = ____ sq ft. Area of all 7 tiles = ____ sq ft.
12. Create your own tile with a different lean but the **same base and height**. Does its area change? Explain using cut-and-slide.

Reflection & Presentation

13. Title your poster "Build It in Detroit." Write 3–4 sentences: Where did you see parallelograms today? Which measurement is always the height? Where else in Detroit could a designer use parallelograms?

Scoring Rubric (20 points)

Category	4 pts	2 pts	0 pts
Accurate drawings (ruler, scale, labels)	All shapes neat & to scale	Some errors	Missing
Heights marked (dotted line + right-angle box)	Every shape	Some shapes	None
Area calculations (Jobs 1-3)	All correct, work shown	Some correct	Missing
Color & design	Clear patterns, colored neatly	Partly colored	Not colored
Reflection sentences	3-4 thoughtful sentences	1-2 sentences	Missing

Teacher Answer Key

3. $10.5 \times 18 = 189$ sq ft ($105 \times 18 = 1,890 \rightarrow$ one decimal place). 3 stalls: 567 sq ft. **4.** $9 \times 18 = 162$ sq ft; the angled stall uses $189 - 162 = 27$ sq ft more (a trade-off for easier turning). **5.** 20.8 ft is the slanted stripe, not the height; area = base $\times$ height = 10.5×18 . **7.** Box $36 - 4$ triangles $\times 4.5 = 36 - 18 = 18$ grid squares. **8.** $90 \times 90 = 8,100$ sq ft. **9.** Yes: both pairs of opposite sides are parallel (and equal); a square is a special parallelogram. **11.** $4 \times 3 = 12$ sq ft; $7 \times 12 = 84$ sq ft. **12.** No — same base and same height give the same area; cutting the triangle off one end and sliding it to the other makes the same rectangle.

Teacher note on the Detroit skyscraper idea: we looked for a downtown Detroit skyscraper whose top is built from parallelograms and could not confirm one. Ally Detroit Center's crown uses paired neo-Gothic spires, and the Hudson's Detroit tower steps back as it rises. The Guardian Building's geometric patterns and the angled-parking and baseball-diamond examples are used instead because they can be checked.

Sources

- [Indiana DOT Design Manual, Fig. 51-4A — Parking angle dimensions](#)
- [Guardian Building \(Wikipedia\)](#)
- [Ally Detroit Center \(Wikipedia\)](#)
- [Dezeen — SHoP Architects' Hudson's Detroit skyscraper](#)